

Credit Risk Model Results

Beta-Binomial and Gaussian Threshold Models

1 Beta-Binomial Model

Table 1: Beta-Binomial model parameters by portfolio

Portfolio	m	Total EAD	p	EAD-weighted LGD	Observed variance	Independent variance	ρ_y	s	α	β
Portfolio 1	28	631,058,707	1.99%	45.81%	0.00078608	0.00071566	0.37%	265.7617	5.2880	260.4737
Portfolio 2	28	704,450,945	3.52%	49.17%	0.00150996	0.00124632	0.81%	123.0921	4.3330	118.7591
Portfolio 3	35	769,806,387	3.12%	47.88%	0.00092319	0.00088199	0.14%	710.7209	22.1605	688.5604

2 Gaussian One-Factor Threshold Model

Table 2: Gaussian one-factor threshold model parameters by portfolio

Portfolio	m	Total EAD	p	Threshold d	ρ_y	ρ	Asset corr. ρ^2
Portfolio 1	28	631,058,707	1.99%	-2.055868	0.37%	17.19%	2.96%
Portfolio 2	28	704,450,945	3.52%	-1.809311	0.81%	20.59%	4.24%
Portfolio 3	35	769,806,387	3.12%	-1.863723	0.14%	9.21%	0.85%

3 Final Model Comparison

Table 3: Final comparison: Beta-Binomial vs. Gaussian one-factor model

Model	Portfolio	Method	Expected Loss	VaR 99%	VaR 99.9%	ES 99%	ES 99.9%
Beta-Binomial	Portfolio 1	Simulation	0.91%	4.91%	6.54%	5.23%	6.81%
Beta-Binomial	Portfolio 1	Analytical	0.91%	4.91%	6.54%	5.25%	6.84%
Beta-Binomial	Portfolio 2	Simulation	1.72%	7.02%	10.54%	7.65%	10.96%
Beta-Binomial	Portfolio 2	Analytical	1.73%	7.02%	10.54%	7.66%	11.04%
Beta-Binomial	Portfolio 3	Simulation	1.49%	5.47%	8.21%	5.85%	8.45%
Beta-Binomial	Portfolio 3	Analytical	1.49%	5.47%	8.21%	5.83%	8.46%
Gaussian One-Factor	Portfolio 1	Simulation	0.92%	5.54%	7.99%	6.62%	8.99%
Gaussian One-Factor	Portfolio 1	Analytical	0.91%	2.13%	2.79%	2.41%	3.07%
Gaussian One-Factor	Portfolio 2	Simulation	1.73%	8.09%	11.37%	9.51%	13.02%
Gaussian One-Factor	Portfolio 2	Analytical	1.73%	4.28%	5.67%	4.88%	6.27%
Gaussian One-Factor	Portfolio 3	Simulation	1.49%	6.31%	8.47%	7.26%	9.52%
Gaussian One-Factor	Portfolio 3	Analytical	1.49%	2.34%	2.70%	2.50%	2.84%